Jenkins Installation and Configuration SOP

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1. Jenkins Installation and Configuration

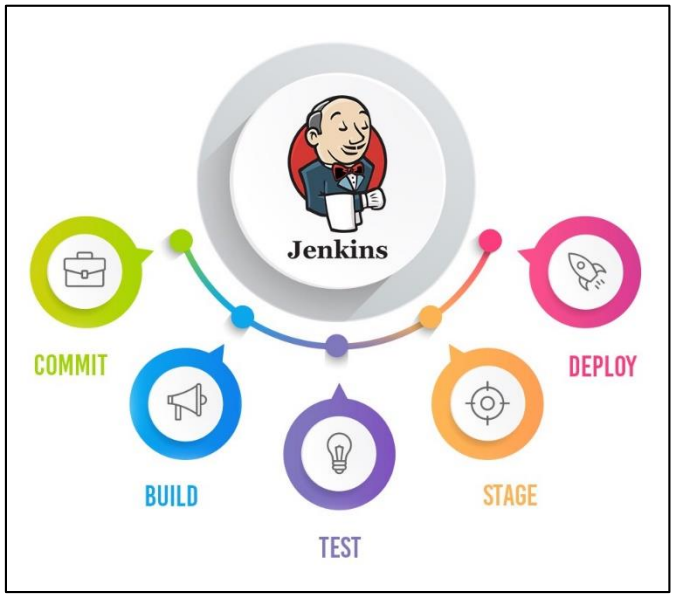
1.1 Description

Jenkins® is an open source automation server. With Jenkins, organizations can accelerate the software development process by automating it. Jenkins manages and controls software delivery processes throughout the entire lifecycle, including build, document, test, package, stage, deployment, static code analysis and much more.

We can set up Jenkins to watch for any code changes in places like GitHub, Bitbucket or GitLab and automatically do a build with tools like Maven and Gradle. We can utilize container technology such as Docker and Kubernetes, initiate tests and then take actions like rolling back or rolling forward in production.

1.2 Architecture Diagram

The diagram below displays a visual representation of the underlying application architecture:



1.3 Lab Steps

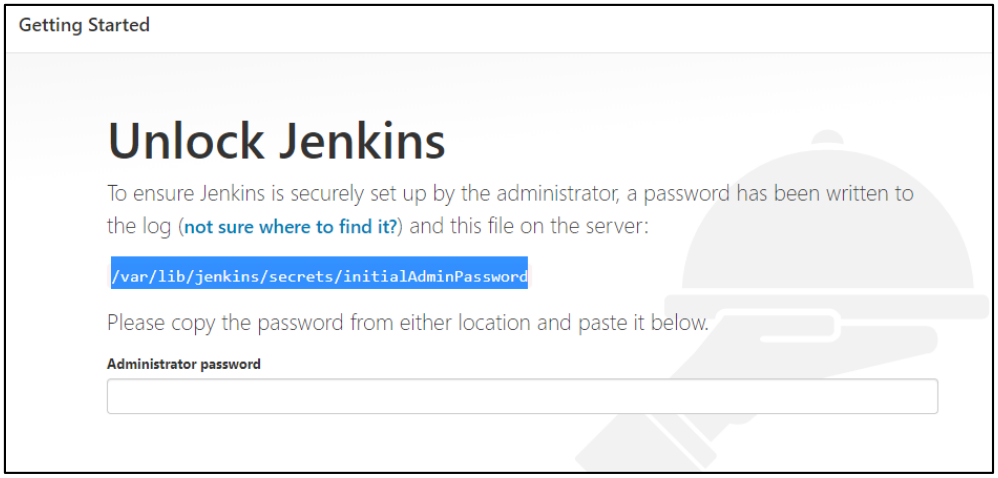
Jenkins is a well-known open-source continuous integration and continuous development automation tool. Jenkins Pipeline is a combination of plugins that support the integration and implementation of continuous delivery pipelines using Jenkins.
Follow the steps outlined below to achieve the objectives of this lab exercise:

Prerequisites:

- Sign in to the AWS Management Console.
- An Amazon EC2 key pair, if you do not have one.

1. Install the Jenkins and docker in AWS machine and execute the build steps
    a. Execute the following steps to install the Jenkins in Centos 7 machine:
        i. Once you open your AWS machine or any other cloud instance get into root mode.

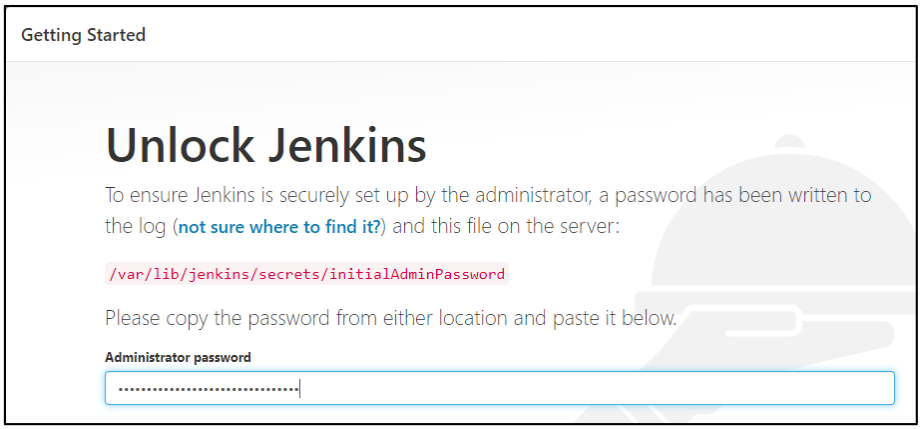
        ii. Enter the following address from your web browser to test Jenkins:
        http://< your_public_ip_address >:8080
        Once you enter the browser, the below screen displays:



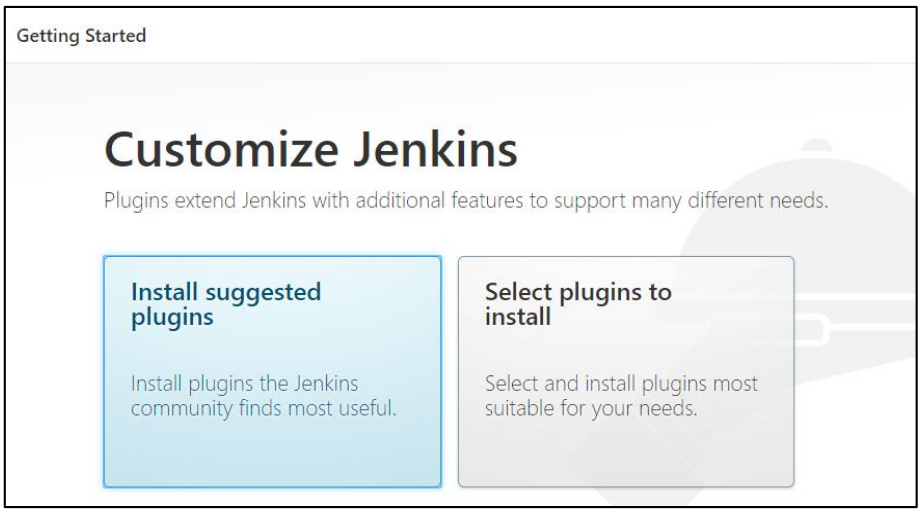
        iii. Type the following command in your Jenkins server and get the password and paste it in the Administrator password section in the above screen:

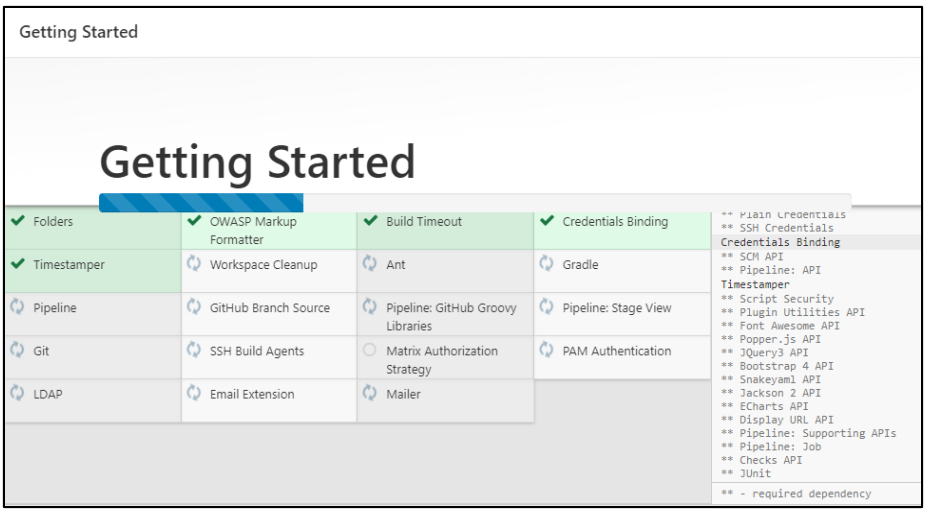
```
cat /var/lib/jenkins/secrets/initialAdminPassword
```

```
root@centos-jenkins ~]# cat /var/lib/jenkins/secrets/initialAdminPassword
2a2af0961044a57860dd977629db501
root@centos-jenkins ~]#
```

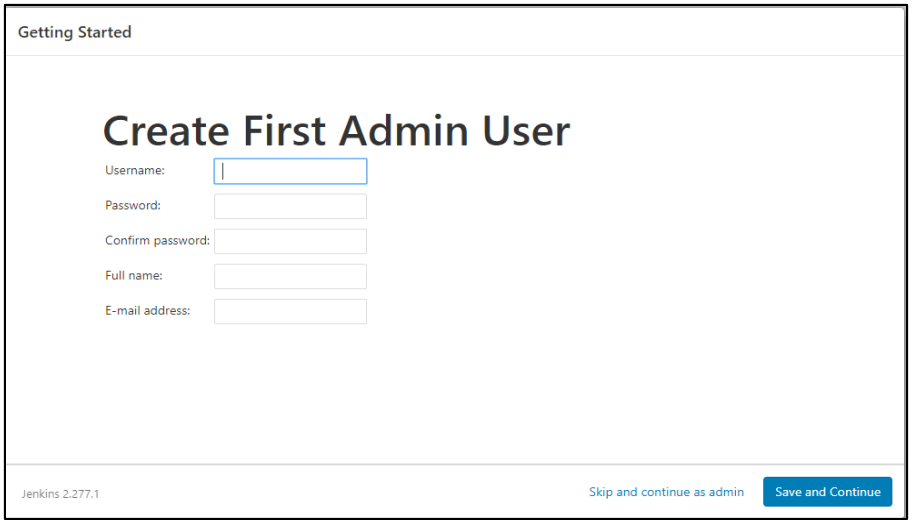


iv. Select the **Install suggested plugins** option:

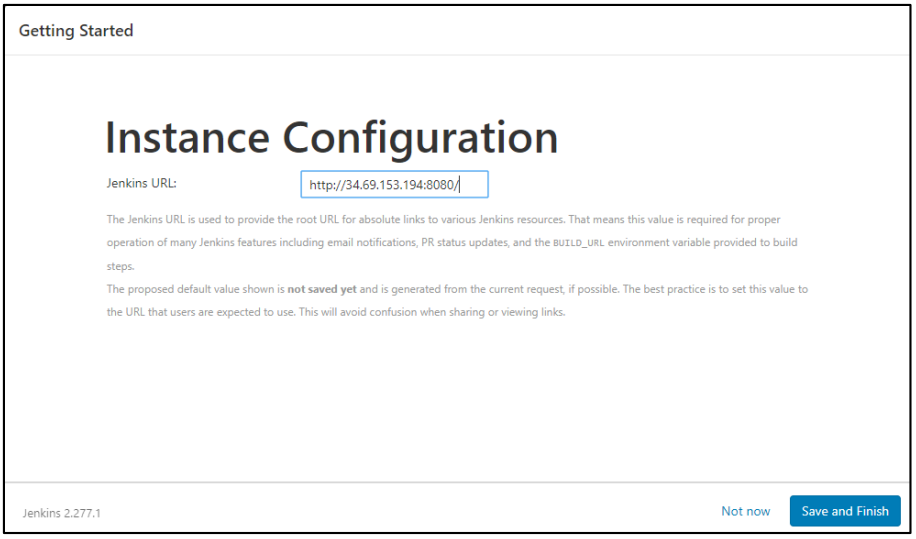


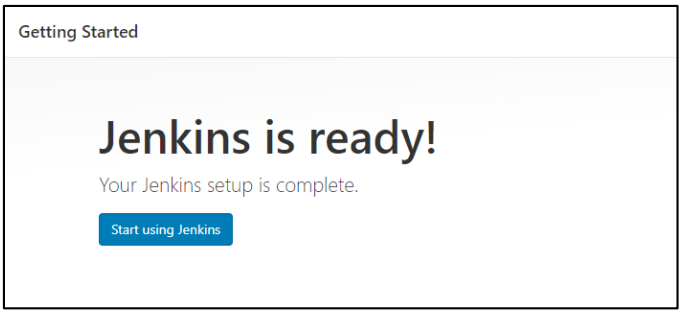


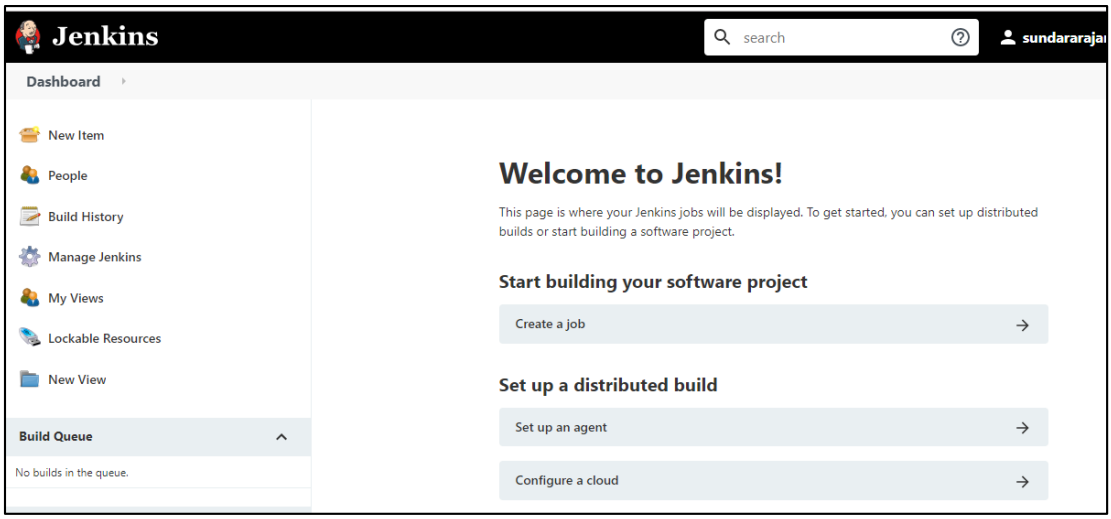
v. Click **Save and Continue** once you provided the required info in the screen:



vi. Click **Save and Finish** button:







1.4 Troubleshooting

S. No	Problem	Solution
1	Jenkins not able to launch it in the **port 8080** and receiving **page not found** error.	Open the firewall for specified port in cloud environment.

1.5 Supporting References

Refer the below link for additional information:

1. https://medium.com/appgambit/build-pipeline-with-jenkins-c13bf788a3f0